N-GRAM ACTIVITY SHEET

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Sentence: - the cat sat on the mat. the dog sat on the rug . the cat ate fish .

ACTIVITY 1: BACKOFF (penalty = 0.4)

Using backoff with a penalty of 0.4, find the score for each phrase. Say which level (trigram, bigram or unigram) you used.

a) "the cat sat"

The trigram "the cat sat" occurs once, and the context "the cat" occurs twice. Therefore, use the trigram level.

P(sat | the cat) = C(the cat sat) / C(the cat) = 1 / 2 = 0.5

Final answer: Score = 0.5; Level = Trigram.

b) "dog ate fish"

The trigram "dog ate fish" does not occur. The bigram "ate fish" occurs once, and "ate" occurs once. Back off to the bigram level.

P(fish | ate) = C(ate fish) / C(ate) = 1 / 1 = 1

Score = 1 × 0.4 = 0.4

Final answer: Score = 0.4; Level = Bigram (backoff).

c) "dog sat fish"

The trigram "dog sat fish" does not occur in the corpus. The bigram "sat fish" also does not occur. Therefore, back off to the unigram level for "fish".

P(fish) = C(fish) / N = 1 / 19

Since we back off twice, apply the penalty 0.4 twice:

Score = $0.4^2 \times (1 / 19) = 0.16 / 19 \approx 0.00842$

Final answer: Score ≈ 0.00842; Level = Unigram (after two backoff steps).

Assumption: The penalty is multiplied once for each backoff step.

ACTIVITY 2: DELETED INTERPOLATION (λ1 + λ2 + λ3 = 1)

Part A

Question: Using λ1 = 0.1, λ2 = 0.3, λ3 = 0.6, find P(sat | the cat).

The deleted interpolation formula is:

P(w | u, v) = λ1 P(w) + λ2 P(w | v) + λ3 P(w | u, v)

Here, w = sat, v = cat, and u = the.

Step 1: Calculate unigram probability.

P(sat) = 2 / 19 = 0.10526

Step 2: Calculate bigram probability.

P(sat | cat) = C(cat sat) / C(cat) = 1 / 2 = 0.5

Step 3: Calculate trigram probability.

P(sat | the cat) = C(the cat sat) / C(the cat) = 1 / 2 = 0.5

Step 4: Apply interpolation.

P(sat | the cat) = 0.1(0.10526) + 0.3(0.5) + 0.6(0.5)

= 0.010526 + 0.15 + 0.30 = 0.460526

Final answer: P(sat | the cat) ≈ 0.4605.

Part B

Question: Compare Set 1, λ = (0.1, 0.3, 0.6), and Set 2, λ = (0.3, 0.6, 0.1), for the held-out sentence "the dog ate fish". The trigrams are "the dog ate" and "dog ate fish".

The interpolation formula is:

P(w | u, v) = λ1 P(w) + λ2 P(w | v) + λ3 P(w | u, v)

Step 1: Calculate the probability of "the dog ate".

Here, the target word is "ate".

Unigram probability: P(ate) = 1 / 19

Bigram probability: P(ate | dog) = 0

Trigram probability: P(ate | the dog) = 0

$P_1 = 0.1(1 / 19) + 0.3(0) + 0.6(0) = 0.005263$

$P_2 = 0.3(1 / 19) + 0.6(0) + 0.1(0) = 0.015789$

Step 2: Calculate the probability of "dog ate fish".

Here, the target word is "fish".

Unigram probability: P(fish) = 1 / 19

Bigram probability: P(fish | ate) = 1

Trigram probability: P(fish | dog ate) = 0

$P_1 = 0.1(1 / 19) + 0.3(1) + 0.6(0) = 0.305263$

$P_2 = 0.3(1 / 19) + 0.6(1) + 0.1(0) = 0.615789$

Step 3: Add the natural logarithms.

Total log probability = Σ ln P(word | context)

$\ln P_1 = \ln(0.005263) + \ln(0.305263)$

$\approx -5.2470 + (-1.1866) = -6.4336$

$\ln P_2 = \ln(0.015789) + \ln(0.615789)$

$\approx -4.1484 + (-0.4849) = -4.6333$

Step 4: Final decision and justification.

Deleted interpolation chooses Set 2 because its total log probability (−4.6333) is higher than Set 1's (−6.4336). A higher log probability indicates that the held-out sentence receives a higher probability under the model.

Selected weights: $\lambda = (0.3, 0.6, 0.1)$.

Conclusion: Set 2 gives the better score for the held-out sentence because it assigns more weight to the unigram and bigram probabilities, which provide useful information when the corresponding trigram is unseen.

ACTIVITY 3: ENTROPY AND PERPLEXITY

Formula: $H = -\Sigma\, p_i \log_2 p_i$

Perplexity: $PP = 2^H$

Part A

Question: Weather: Sunny p = 0.5, Cloudy p = 0.25, Rainy p = 0.25. Find the entropy and perplexity.

Step 1: Calculate entropy.

$H = -[0.5 \log_2(0.5) + 0.25 \log_2(0.25) + 0.25 \log_2(0.25)]$

Since $\log_2(0.5) = -1$ and $\log_2(0.25) = -2$:

$H = -[0.5(-1) + 0.25(-2) + 0.25(-2)]$

$H = -[-0.5 - 0.5 - 0.5] = 1.5$ bits

Step 2: Calculate perplexity.

$PP = 2^H = 2^{1.5} \approx 2.828$

Final answer: Entropy = 1.5 bits; Perplexity ≈ 2.828.

Part B

Question: List the words that come after "the" and their probabilities. Find the entropy and perplexity of guessing the next word after "the". What is the entropy of the next word after "sat"?

Step 1: List the words that come after "the".

The word "the" occurs 5 times in the corpus.

Word after "the"	Frequency	Probability
cat	2	2/5 = 0.4
mat	1	1/5 = 0.2
dog	1	1/5 = 0.2
rug	1	1/5 = 0.2
Total	5	1.0

Step 2: Calculate entropy.

$H = -[0.4 \log_2(0.4) + 0.2 \log_2(0.2) + 0.2 \log_2(0.2) + 0.2 \log_2(0.2)]$

Using $\log_2(0.4) \approx -1.3219$ and $\log_2(0.2) \approx -2.3219$:

$H = -[0.4(-1.3219) + 3(0.2)(-2.3219)]$

$H = -[-0.5288 - 1.3932] \approx 1.9219$ bits

Step 3: Calculate perplexity.

$PP = 2^H = 2^{1.9219} \approx 3.789$

Step 4: Find the entropy of the next word after "sat".

In the corpus, "sat" occurs twice, and both times the next word is "on".

$P(\text{on} \mid \text{sat}) = 2 / 2 = 1$

$H = -1 \log_2(1) = 0$ bits

Reason: The next word after "sat" is always "on" in this corpus, so there is no uncertainty.

Final answers:

- Part A entropy = 1.5 bits
- Part A perplexity ≈ 2.828
- Part B entropy after "the" ≈ 1.9219 bits
- Part B perplexity after "the" ≈ 3.789
- Part B entropy after "sat" = 0 bits